University of Bozen-Bolzano
Bachelor's Degree in Informatics and Management of Digital Business

Economics of Digital Markets

Problem set #3

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In this simplified ultimatum game, the Proposer offers the Responder either 2 euros or 5 euros from a 10-euro pie. The 2-euro offer (low offer) is denoted by L, while the 5-euro offer (fair offer) is denoted by F. The Responder sees the offer and accepts (A) or rejects (R). Acceptance gives each player their proposed share; rejection gives both zero. The sequential game is summarized in the tree below.

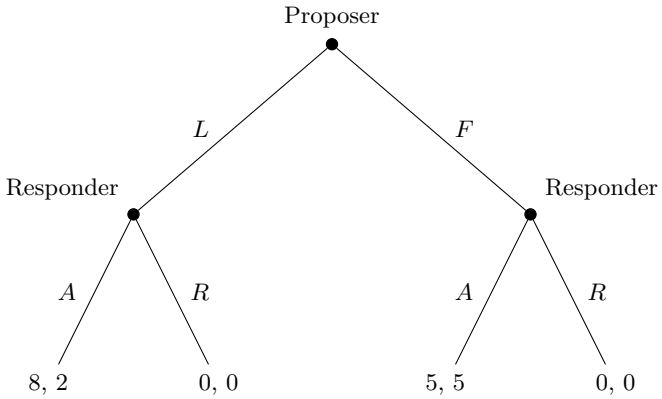


Figure 1: *The ultimatum game as a game tree.*

In this setting, a strategy for the Responder is a contingent plan that specifies a choice *after each possible offer*, even though only one offer will actually be made. In the two-letter strategies below, the first letter specifies the Responder's choice if the Proposer chooses L, and the second specifies their choice if the Proposer chooses F. For example, AR means accept L and reject F.

		Responder			
		AA	AR	RA	RR
Proposer	L	8, 2	8, 2	0, 0	0, 0
	F	5, 5	0, 0	5, 5	0, 0

Figure 2: *The same game as a payoff matrix.*

Answer the following questions.

(a) Using the matrix, identify the Proposer's best response(s) to each of the Responder's strategies and the Responder's best response(s) to each of the Proposer's strategies. Then find all Nash equilibria and explain why each is an equilibrium.

(b) Suppose the Proposer has already offered 2 euros. Which action is optimal for

the Responder, and why? What would be optimal if the Proposer had offered 5 euros instead?

(c) If the players reason sequentially, which of the Nash equilibria from part (a) would they select? *(Hint: Start from each subtree in which the Responder makes a decision, then work back to the Proposer's initial decision.)*

(d) Now suppose that the Proposer may offer any amount y from 0 to 10 euros, in increments of one cent. An equal split (5 euros each) is the social norm. The Responder's utility from accepting is y. For an offer below 5 euros, the Responder's utility from rejecting is $5 - y$, reflecting satisfaction from punishing the Proposer for making an offer below the norm. Offers of at least 5 euros are accepted, and the Responder accepts when indifferent. Calculate the Responder's minimum acceptable offer. Would the 2-euro offer be accepted under these assumptions?

(e) In questions (a) to (d), we assumed that players cared only about their own monetary payoffs. Is this a reasonable assumption in the ultimatum game? Why?